

FLEET MOBILITY & RELATIONAL DATA TRANSFORMATION

The Vehicle Inspection Application (VIA) Case Study



LEAD SPECIALIST David Liong

- Professional Scrum Master (PSM I)
- Power Platform Developer Focus
- Solution Architect PL-600 Focus
- Experienced Technical BA & Lead



TECH STACK

Power Platform Core

- Microsoft Dataverse (3NF Schema)
- Power Apps (Canvas & Model-Driven)
- Power Automate Flow Orchestration
- SharePoint Online Document Library



CLIENT CONTEXT

RAC Motoring Services

- Sector: Automotive & Roadside
- Scope: 200,000 annual inspections
- Goal: Process digitisation
- Impact: Zero-downtime go-live



Governed & Scalable
Built on Microsoft Power Platform
and Dataverse.



Un-Governed Legacy Paper Workflows Constrained Growth

Pillar 1: THE EXECUTIVE CHALLENGE (THE PROBLEM)

Executive Impact

RAC's mobile patrols and auto mechanics were bound to a slow, physical paper workflow for car inspections. The lack of standard automation and real-time validation meant that simple transcription took up valuable frontline capacity, exposing the organization to operational friction and compliance threats.



OPERATIONAL INEFFICIENCY



10,000 Hours Wasted Annually

Manual transcription of complex VIN and Engine numbers consumed 2 to 3 minutes of frontline patrol time per job across 200,000 annual runs.

COMPLIANCE & SECURITY THREAT



Field Exposure of Sensitive PII

Physical paper logs with customer and vehicle details remained in roadside vans without obfuscation, creating serious data sovereignty and security risks.

SYSTEM ISOLATION



Siloed Data & Zero Analytics

Data was physically trapped on paper, preventing any real-time tracking, compliance auditing, automated notifications, or Power BI reporting.

BUSINESS RISK

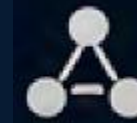


Operational Leakage

Mobile patrols were bound to slow physical forms without standard automation or real-time validation.



Structured Agile Lifecycle Delivery & Mitigated Risks



Process Automation

Power Automate reduces manual effort and errors.

Pillar 2: THE ENGINEERING CRAFT (SKILLS & METHODOLOGY)



1. ANALYSING REQUIREMENTS

Stakeholder Workshop & SID Elicitation

- Facilitated inception workshops directly with frontline patrols to map operational workflows.
- Translated fluid business 'pain' into structured technical and functional requirements.
- Authored comprehensive Solution Intent Documents (SIDs) as the single source of truth.
- Managed technical trade-offs, prioritizing immediate MVP features over long-term API extensions.



2. SCRUM LEADERSHIP & QA

Agile Execution & Zephyr Testing

- Operated as Scrum Master (PSM I), facilitating retrospectives, daily scrums, and backlog grooming.
- Developed and designed a **mobile response canvas app** and coordinated peer reviews with secondary developers.
- Authored QA checklists and test scripts managed in the Zephyr Scale tool for automated testing.
- Fix bug and deploy solutions to coordinate UAT testing by end-users to sign-off for production release.



3. ALM DEPLOYMENT & POST SUPPORT

Environment Staging & Helpdesk Handover

- Published user manual documentations and conducted end-user training.
- Created deployment guides to manage environment transition.
- Produced approved Service Definition and Transition checklists for BAU operations.
- Conducted hand over Level 1 support to the Helpdesk team.

Dataverse Normalisation Strategy

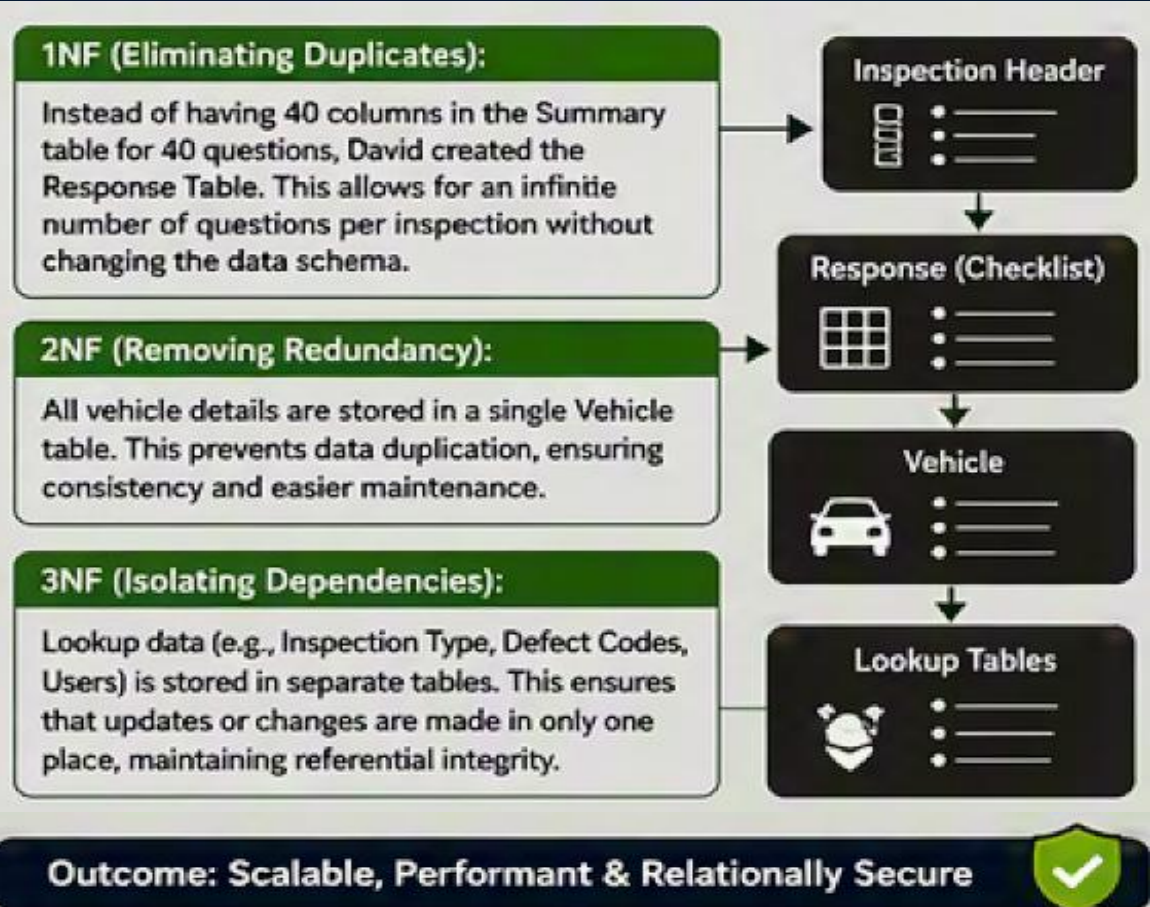


Single Source of Truth

Relational data model ensures accuracy and consistency.

Pillar 3: THE ARCHITECTURAL BLUEPRINT (THE SOLUTION)

Architect the VIA project using this *normalised 5 **Dataverse tables** relational model to ensure the **VIA product could scale to 200,000+ records without performance degradation**. By separating **Questions** from **Responses table**, David made the system 'Future-Proof.' If a user needs to add a new checklist question tomorrow, they don't need a developer to change the UI; they just add a row to the **Questions Table**. This reduces technical debt and ensures the **new question added in question table is auto populated into the canvas app**.



PLATFORM GOVERNANCE & SECURITY

Hardened Security Staging

Custom Dataverse Role (Basic User - MWPSOL-141):

Followed a 'Least Privilege' model - Created a custom role called 'Basic User - MWPSOL-141' by cloning the standard base role and hardening it. This restricted the **Delete** and **Share** permissions on the Inspection tables to protect PII. This ensured that only members of the verified AD Security Group could interact with the data, which is the same level of compliance rigor I'd apply to the client's SaaS environment.

Question & Inspection Summary Table:

Granted **User-level** access of **Create & write** for users who own the records they create are assigned to the basic user security role. The organisation (every other user else in the basic user security role) will have read access to every record, but no one is allowed to delete the records. This ensured that a patrol member could create and edit their own inspections but couldn't necessarily delete records or view inspections from unrelated departments.

Dataverse Normalisation Strategy

Pillar 3.1 : THE ARCHITECTURAL BLUEPRINT (THE SOLUTION)



SECURE
Built for Compliance
and Data Sovereignty



SCALABLE
Designed to Grow
with the Business

PLATFORM GOVERNANCE & SECURITY

Hardened Security Staging

Response Table:

Grant **User-level** access of **Create** for users who **own the records they create**. However, every other user can read any response records and continue to update a record to continue from where the last person had saved it.

Photo Table:

Grant **User-level** access for **Create** for users who **own the records they create**. Admin (organisation) has full visibility to entire record to read, write and delete.

Security Role: Basic User - MWPSOL-141 User

Details

Business unit: org66b3d41

When role is assigned to a Team

Team member gets all team privileges by default.
Team members can inherit team privileges directly based on access level. [Learn more](#)

Member's privilege inheritance

Direct User (Basic) access level and Team privileges

Tables Miscellaneous privileges Privacy-related privileges

Show only assigned tables

Compact Grid View

Table 1	Name	Record ownership	Permission Settings	Create	Read	Write	Delete	Append	Append to	Assign	Share
Custom Tables (3)											
Questions	shmg_questions	User or Team	Reference	None	Organization	None	None	None	None	None	None
Responses	shmg_responses	User or Team	Custom	User	Organization	Organization	None	None	None	None	None
Vehicles	shmg_vehicles	User or Team	Reference	None	Organization	None	None	None	None	None	None

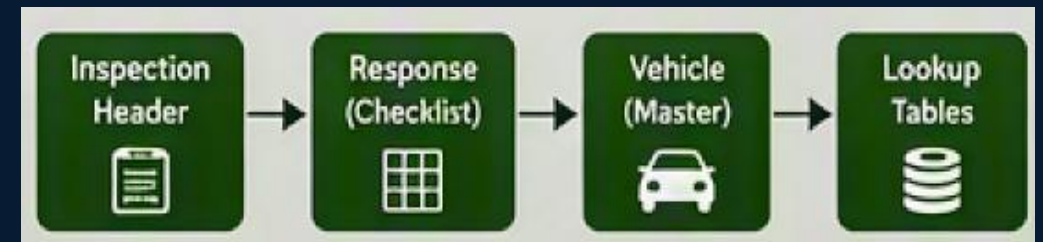
PLATFORM COMPLIANCE SECURITY

Zero Tech Debt

- Relational 5-table Dataverse schema successfully cleared all corporate security and compliance audits.
- Hardened security roles eliminated sensitive PII exposure & protect sensitive data in the field.

Secure Relational 3NF Schema Replaced Flat-File Limits

Pillar 3.2 : THE ARCHITECTURAL BLUEPRINT (THE SOLUTION)



DATABASE NORMALISATION (3NF)

Relational 5-Table Schema

Vehicle (Master Lookup):

Stores persistent asset data (VIN, Rego, Make, Model) to eliminate repetitive typing.

Questions (Metadata Config):

Keeps the system fully data-driven. Checklist questions are loaded from database dynamically without redeploying code.

Summary (Header Transaction):

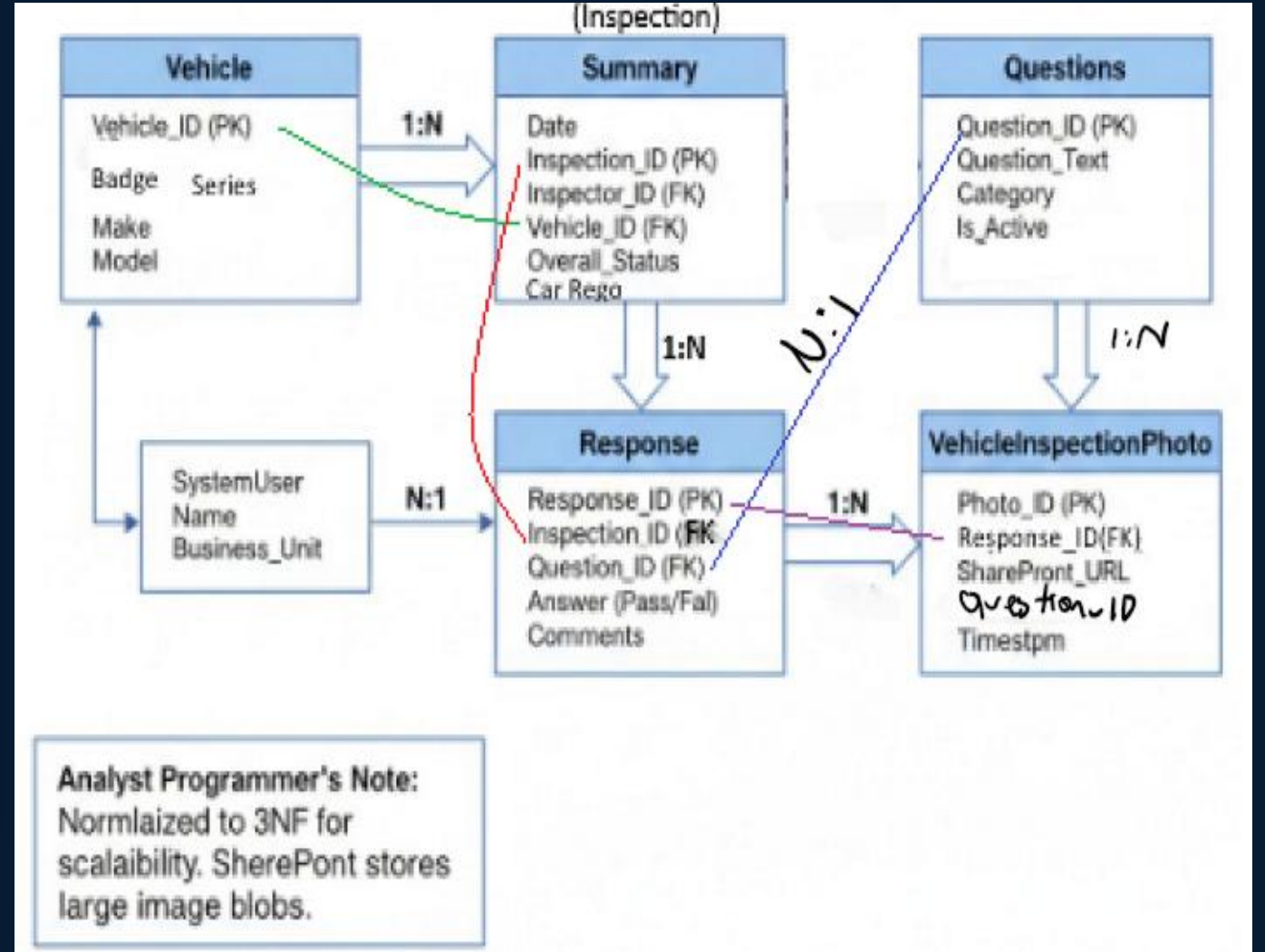
Stores core event details (Inspector name, Overall status, GPS location, timestamp).

Response (Line Items Child):

Captures transactional pass/fail answers for 40+ inspection points linked back to Header.

Media Metadata (Attachments):

Stores secure photo links. Files are offloaded to a governed SharePoint Document Library to reduce Dataverse storage costs.



Operational & Compliance Transformation Yielded 100% Success

Pillar 4: DELIVERING DIRECT, MEASURABLE BUSINESS ROI AND FINANCIAL SAVINGS WITH OPERATIONAL EXCELLENCE



EFFICIENCY UNLOCKED

Digitised inspections save 2-3 minutes per job, reclaiming ~10,000 frontline hours annually.



SECURE & COMPLIANT

PII data secured in Dataverse with role-based access, encryption, and audit trails.



REAL-TIME INSIGHT

Live dashboards and reports enable proactive decision-making and continuous improvement.



FUTURE-READY PLATFORM

Scalable, low-code architecture ready to integrate APIs and future capabilities.

KEY OUTCOMES



200,000+
Annual
Inspections
Digitised



10,000+
Hours
Reclaimed
Annually



Zero
Data Breach
& Security
Incidents



Zero
Downtime
Go-Live
Achieved



Seamless Collaboration

SharePoint document storage and secure sharing.

COMMERCIAL RETURN ON INVESTMENT

AU\$500,000+

RECLAIMED FRONT-LINE CAPACITY

- Directly saved 2 to 3 minutes of manual transcription effort per job.
- Reclaimed the equivalent productivity of over 5 full-time employees (FTEs)
- Eliminated transcription error logs and automated field reporting workflows.

- Direct operational capacity and human labor value returned to Motoring Services.
- Substantially reduced downstream administrative and data remediation costs.
- Established a secure, scalable model for rapid multi-branch system rollout.

Operational & Compliance Transformation Yielded 100% Success

Pillar 4.1 : DELIVERING DIRECT, MEASURABLE BUSINESS ROI AND FINANICAL SAVINGS WITH OPERATIONAL EXCELLENCE

RECLAIMED CAPACITY

Equivalent to approximately 277 to 416 twenty-four-hour days.

QUALITY & REPORTING

Manual paper transcription were eliminated, and field reporting was automated.

SCALE & COMPLIANCE

Five-table Dataverse schema, governed PII and a repeatable rollout blueprint support future expansion.

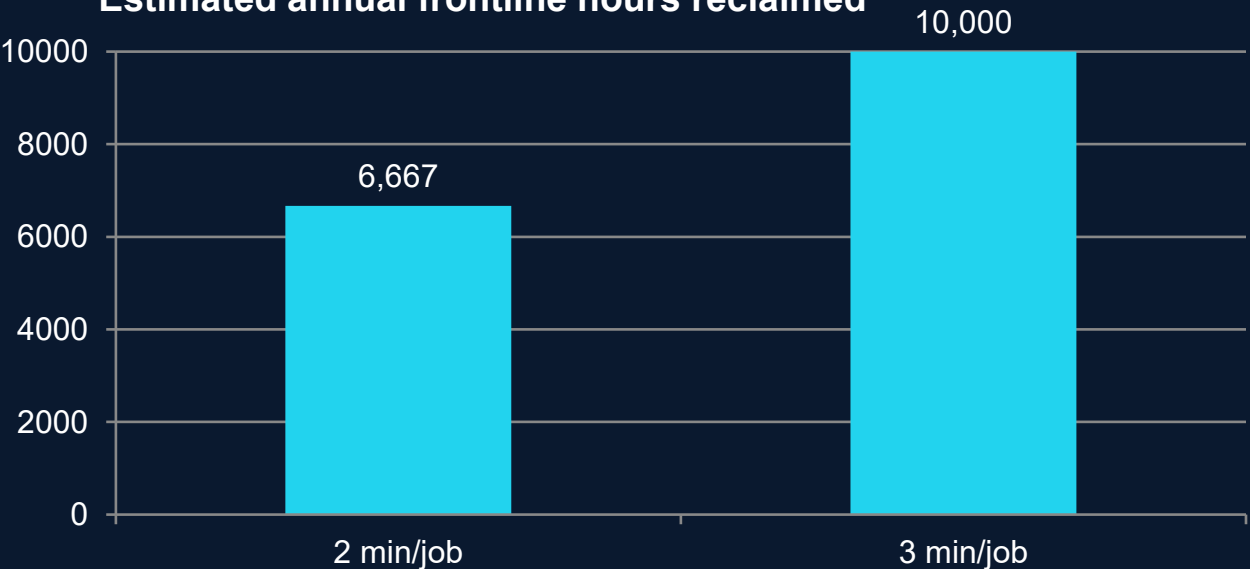
GO-LIVE

3 June 2024



SUSTAINABLE
Lower Costs,
Better Future

Estimated annual frontline hours reclaimed



Why the VIA Matters to Managers

Evidence of David Liong who converts frontline problems into secure, scalable and valuable application.

BUSINESS ANALYSIS

Facilitated frontline workshops, authored SIDs and converted operational pain into a prioritised MVP.

SOLUTION ARCHITECTURE

Designed a secure 3NF Dataverse model, data-driven questions and governed document storage.

DELIVERY LEADERSHIP

Led Scrum, QA, peer review, ALM pipelines, deployment guides and Helpdesk transition.

COMMERCIAL IMPACT

Returned 6,667 to 10,000 hours annually and more than AU\$500K in operational value.



David do not just build apps. He translate requirements, manage trade-offs and solutionise platforms that deliver measurable value.